Non-Profit Support Triage Agent

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Academic Year:2026

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Non-Profit Support Triage Agent

1 Problem Statement & Objectives

1.1 Problem Statement

Non-profit organizations play a significant role in addressing critical social issues such as poverty reduction, disaster response, healthcare support, education initiatives, environmental sustainability, and community welfare programs. These organizations rely heavily on effective communication with donors, volunteers, beneficiaries, and stakeholders. With the rapid expansion of digital communication platforms, non-profit organizations now receive large volumes of incoming messages daily through websites, emails, contact forms, and support portals.

These communications generally include:

- Donation-related issues (payment failures, transaction confirmations, refund requests)
- Volunteer participation inquiries
- Complaints and grievances
- General informational queries
- Urgent crisis-related assistance requests

Traditionally, these communications are processed manually by support teams. Each message is read, interpreted, categorized, and responded to individually. While this method may work for small volumes, it becomes increasingly inefficient as communication traffic grows.

The manual triage process suffers from several major limitations:

1. **Lack of Prioritization**
   Urgent cases may not be identified immediately, resulting in delayed responses to critical matters.
2. **Human Inconsistency**
   Different staff members may interpret urgency and intent differently, leading to inconsistent classification and handling.
3. **Time Inefficiency**
   Manual reading and drafting of responses significantly increases processing time.
4. **Data Underutilization**
   Historical messages contain valuable patterns and insights, but manual systems rarely analyze them systematically.
5. **Scalability Constraints**
   As message volume increases, operational workload increases proportionally, requiring additional staff and resources.

In modern digital environments, non-profit organizations require intelligent systems capable of handling high communication volumes efficiently while maintaining response quality and operational transparency.

Artificial Intelligence (AI), Natural Language Processing (NLP), and Data Analytics provide an opportunity to automate and optimize this communication workflow. By leveraging Large Language Models (LLMs), semantic search techniques, and structured data extraction, organizations can improve prioritization, response time, and data-driven decision-making.

1.2 Project Objectives

The primary objective of this project is to design and implement a comprehensive AI-driven message triage system specifically tailored for non-profit organizational communication workflows. The system aims to integrate Natural Language Processing techniques, Large Language Models, vector database technology, and interactive dashboards to automate message processing while maintaining contextual understanding and operational transparency.

The project aims to design and implement a complete AI-driven message triage system with the following objectives:

Primary Objectives

1. Automatically classify incoming support messages based on urgency levels.
2. Detect intent categories such as donation issues, complaints, or volunteer requests.
3. Extract important structured information using NLP techniques.
4. Generate intelligent, contextual responses.
5. Maintain a searchable vector-based storage system.
6. Provide interactive operational analytics.

Secondary Objectives

- Design a modular and scalable architecture.
- Ensure clean separation of AI agents.
- Enable dashboard-based monitoring.
- Implement vector search using ChromaDB.

1.3 Scope of the Project

In Scope

The system covers:

- Text preprocessing
- LLM-based classification
- Named Entity Recognition
- AI response generation
- Streamlit dashboard interface
- Analytics visualization
- ChromaDB integration

Out of Scope

- Real-time email integration
- Voice-based message processing
- Multilingual deployment
- Enterprise CRM integration

2 2. Proposed Solution

2.1 Overview of Proposed System

The proposed system is an Artificial Intelligence–driven operational intelligence platform designed to automate the triaging of incoming communications for non-profit organizations. The system integrates advanced Natural Language Processing (NLP), Large Language Models (LLMs), vector database technology, and interactive dashboard analytics to streamline message processing, improve response quality, and generate actionable insights.

Non-profit organizations frequently deal with diverse incoming communications, including donation-related queries, volunteer registrations, complaints, support requests, and general inquiries. Manual processing of such communications often leads to delayed responses, inconsistent prioritization, and underutilization of valuable information contained within these messages. The proposed system addresses these challenges by introducing an automated AI pipeline capable of understanding, categorizing, extracting, and responding to messages in real time.

The system employs a modular multi-agent pipeline architecture where each agent performs a specialized task. This modular approach ensures scalability, maintainability, and ease of future enhancements.

Core Pipeline Components:

1. Message Preprocessing
2. Classification Agent
3. Named Entity Recognition Agent
4. Response Generator Agent
5. Orchestrator Logic
6. Vector Database Storage (ChromaDB)
7. Analytics Dashboard Interface

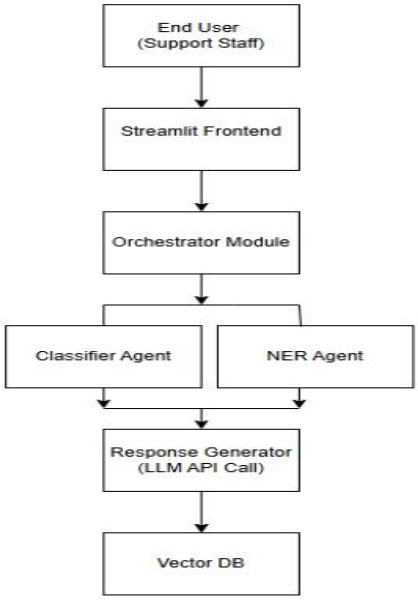
Each component operates independently while contributing to a unified, seamless workflow. This architecture supports efficient data processing, structured output generation, and comprehensive operational analytics.

Communication Processing Workflow

2.2 System Architecture

The system architecture follows a layered design that integrates AI-driven processing with user-facing analytics. The architecture ensures a structured flow of information from raw input to actionable insights.

High-Level Workflow



This workflow demonstrates how raw textual input is transformed into structured intelligence and visual analytics through successive processing stages.

Architectural Layers Description

1. Input Layer

This layer captures incoming user messages through a graphical interface developed using Streamlit. It serves as the entry point for the entire system.

2. Processing Layer

This layer contains multiple AI agents responsible for:

- Text cleaning and normalization
- Message classification
- Entity extraction
- Automated response generation

Each agent is designed to operate independently, ensuring flexibility in system updates.

3. Storage Layer

ChromaDB is used as a vector database to store message embeddings. This enables:

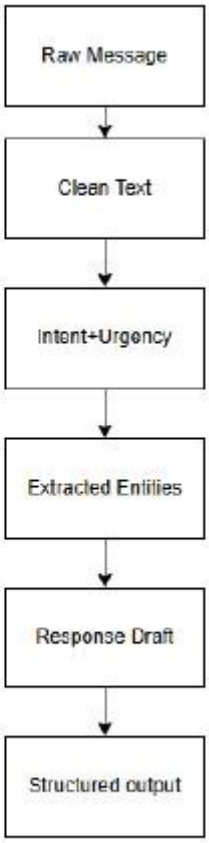
- Semantic similarity search
- Historical message analysis
- Retrieval-Augmented Generation capabilities

4. Presentation Layer

The Streamlit dashboard visualizes:

- Classification results
- Historical trends
- Urgency distributions
- Intent analytics

This enhances operational visibility.



2.3 Architectural Design Philosophy

The system architecture is guided by several important design principles that ensure robustness, scalability, and maintainability.

Modularity

Each AI agent performs a specific function. This modular design allows individual components to be updated or replaced without affecting the overall system.

Scalability

The architecture supports future expansion, such as:

- Sentiment analysis integration
- Multi-language processing
- Advanced predictive analytics

Extensibility

New AI modules can be integrated easily due to the layered structure.

Separation of Concerns

Backend AI processing is completely separated from frontend visualization, improving maintainability and clarity.

Data-Driven Analytics

All processed messages are stored systematically, enabling advanced analytics and operational intelligence.

Principle	Description	Benefits
Modularity	Independent AI agents	Easier maintenance
Scalability	Expandable architecture	Future-proof system
Extensibility	Easy feature addition	Flexible design
Separation of Concerns	Backend/frontend separation	Clean structure
Data-driven Design	Stored outputs	Analytics capability

Table2.1: Architectural Design Principles

3. Technical Implementation

3.1 Preprocessing Module

The preprocessing module is the initial stage of the AI pipeline, responsible for preparing raw textual input for analysis. Since user-generated messages often contain irregular formatting, unnecessary whitespace, or inconsistent punctuation, preprocessing is essential for ensuring reliable AI model performance.

This module performs several cleaning operations to standardize input data.

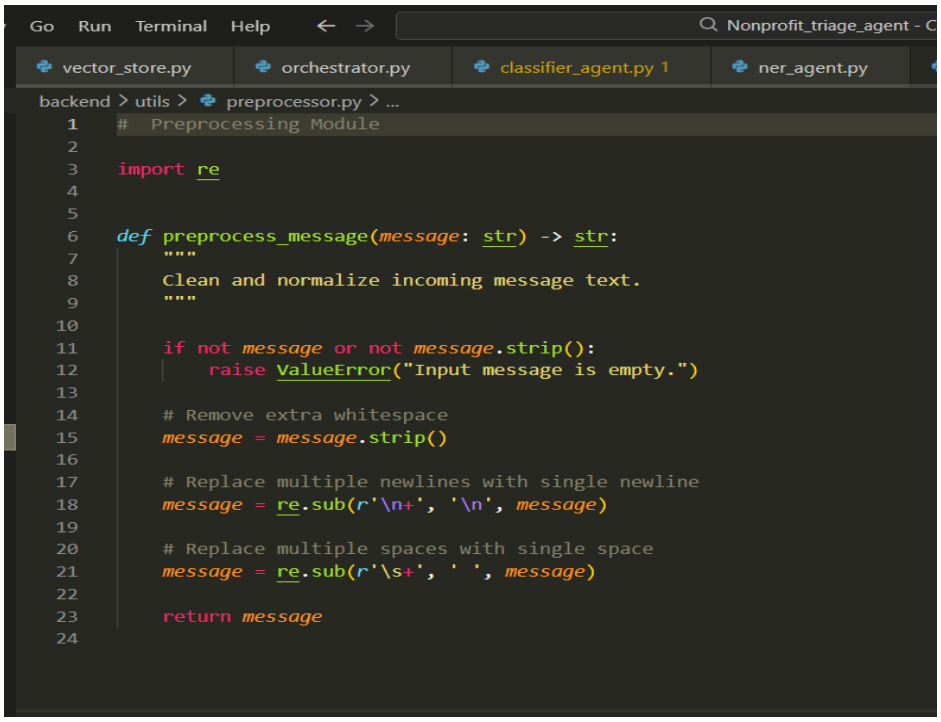
Key Operations Performed:

- Removal of leading and trailing whitespace
- Elimination of extra line breaks
- Normalization of spacing between words
- Standardization of textual format

These steps significantly improve downstream AI processing accuracy.

Clean input ensures:

- Better classification performance
- Improved entity recognition accuracy
- Reduced noise during response generation




```python
#  Preprocessing Module

import re


def preprocess_message(message: str) -> str:
    """
    Clean and normalize incoming message text.
    """

    if not message or not message.strip():
        raise ValueError("Input message is empty.")

    # Remove extra whitespace
    message = message.strip()

    # Replace multiple newlines with single newline
    message = re.sub(r'\n+', '\n', message)

    # Replace multiple spaces with single space
    message = re.sub(r'\s+', ' ', message)

    return message
```

3.2 Classification Agent

The classification agent is one of the most critical components of the system. It utilizes prompt engineering techniques combined with a Large Language Model (Mistral-7B via OpenRouter) to classify incoming messages.

This agent performs two primary tasks:

Urgency Classification

Messages are categorized into:

- High urgency
- Medium urgency
- Low urgency

Urgency classification allows organizations to prioritize critical communications.

Intent Detection

Messages are classified into intent categories such as:

- Donation Issue
- Complaint
- Volunteer Request
- General Inquiry

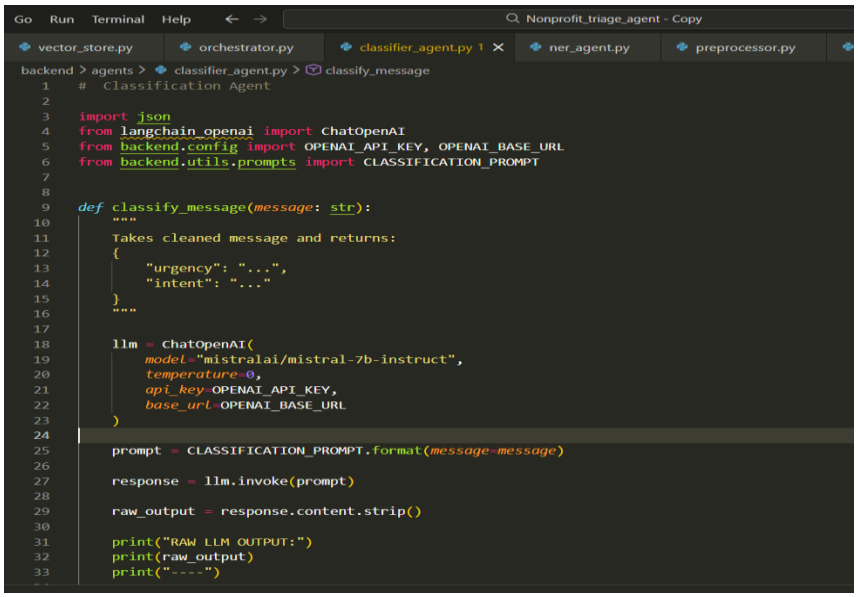
The classification output is structured in JSON format, enabling easy integration with subsequent processing modules.

Importance of Classification

- Enables efficient resource allocation
- Improves response prioritization
- Reduces response delays
- Supports operational decision-making

Category	Description	Example
High Urgency	Immediate attention required	Payment failure
Medium Urgency	Important but not critical	Volunteer inquiry
Low Urgency	General information request	Program details

Table 3.1 : Classification Categories



3.3 Named Entity Recognition (NER)

The Named Entity Recognition module extracts structured information from unstructured text messages. This transformation is essential for converting textual data into analyzable datasets.

Entities Extracted:

- Person Name

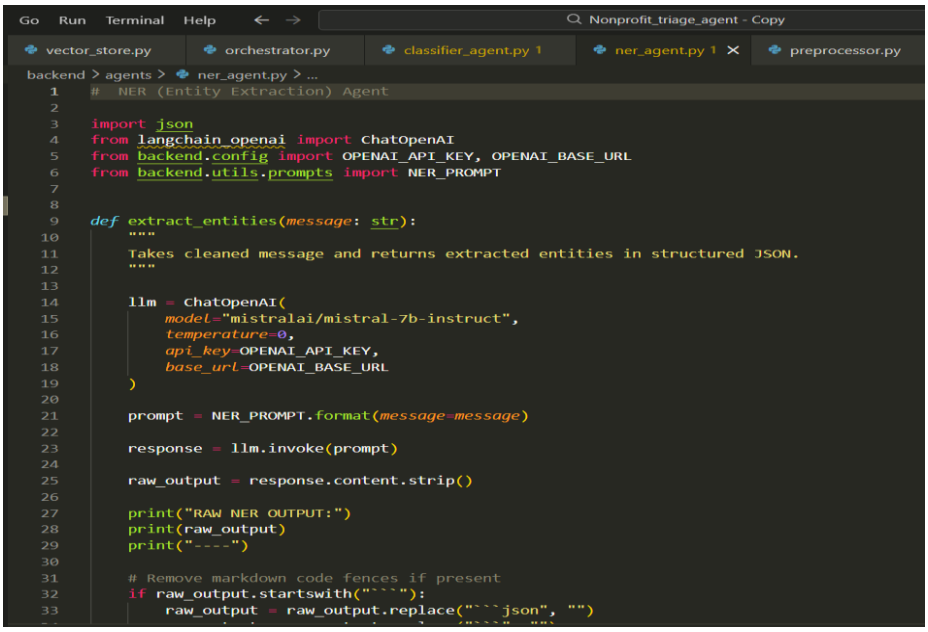
- Donation Amount
- Date
- Organization Name
- Email Address
- NER enables deeper insights into communication patterns and enhances response personalization.

Importance of NER:

- Structured data generation
- Enhanced analytics capability
- Personalized response generation
- Improved operational record keeping

Entity Type	Purpose
Person Name	Donor identification
Amount	Financial analytics
Date	Timeline tracking
Email	Communication follow-up

Table 3.2: Extracted Entities




```python
# NER (Entity Extraction) Agent

import json
from langchain_openai import ChatOpenAI
from backend.config import OPENAI_API_KEY, OPENAI_BASE_URL
from backend.utils.prompts import NER_PROMPT


def extract_entities(message: str):
    """
    Takes cleaned message and returns extracted entities in structured JSON.
    """

    llm = ChatOpenAI(
        model="mistralai/mistral-7b-instruct",
        temperature=0,
        api_key=OPENAI_API_KEY,
        base_url=OPENAI_BASE_URL
    )

    prompt = NER_PROMPT.format(message=message)

    response = llm.invoke(prompt)

    raw_output = response.content.strip()

    print("RAW NER OUTPUT:")
    print(raw_output)
    print("----")

    # Remove markdown code fences if present
    if raw_output.startswith("```"):
        raw_output = raw_output.replace("```json", "")
```

3.4 Response Generator

The response generation module uses LLM capabilities to create contextual, professional responses. It considers both classification results and extracted entities.

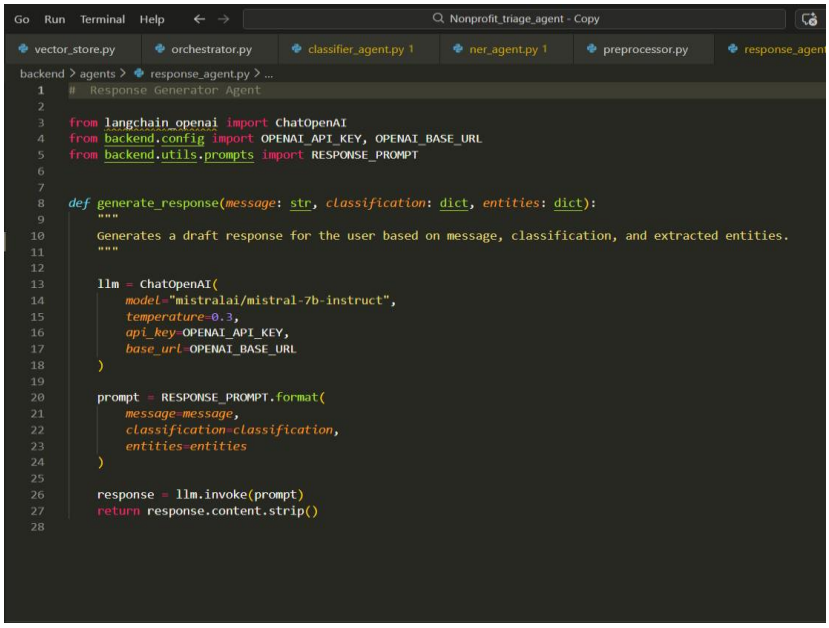
The generated responses are:

- Context-aware
- Personalized
- Professionally structured
- Operationally useful

This significantly reduces manual drafting effort.

Benefits:

- Faster response generation
- Improved donor satisfaction
- Consistent communication tone
- Reduced human workload



3.5 ChromaDB Vector Storage

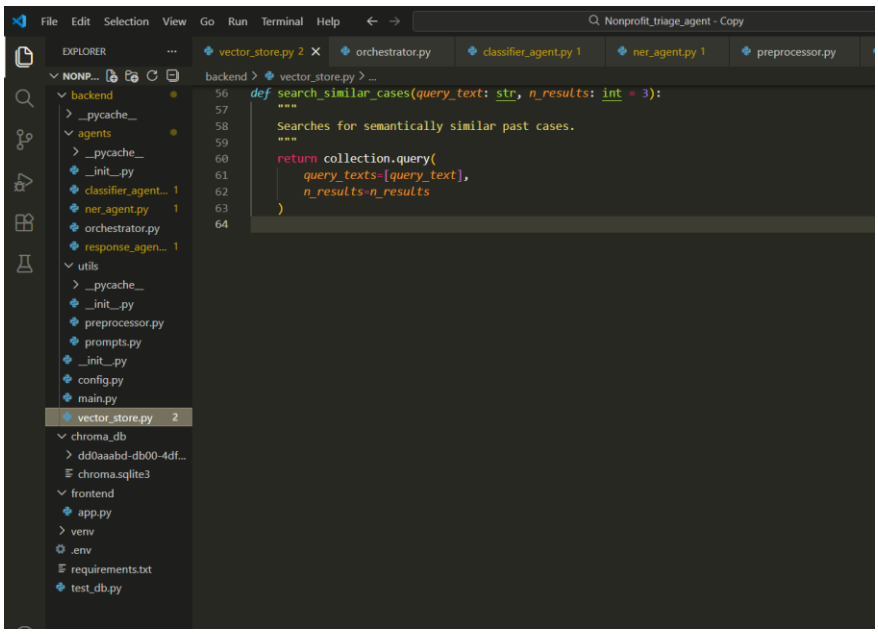
ChromaDB is employed as a vector database to store embeddings of processed messages. Vector storage enables semantic similarity search, which is more effective than traditional keyword search.

This approach supports:

- Similar case retrieval
- Pattern analysis
- Retrieval-Augmented Generation (RAG)
- Contextual recommendations

Advantages of Vector Storage

Advantage	Explanation
Semantic Search	Context-based retrieval
Efficient Indexing	Faster search operations
Scalability	Handles large datasets
AI Integration	Supports future enhancements



4. Dashboard Implementation

The dashboard component of the AI-powered Non-Profit Support Triage Agent plays a critical role in transforming raw AI-generated outputs into actionable operational intelligence. While backend AI agents handle classification, entity extraction, and response generation, the dashboard ensures that these outputs are accessible, interpretable, and useful for decision-making. The dashboard is implemented using Streamlit, which enables rapid development of interactive web-based analytical interfaces.

The dashboard is divided into three major functional sections:

1. Triage Interface
2. History Dashboard
3. Analytics Dashboard

Each section serves a unique purpose while collectively contributing to improved organizational efficiency.

4.1 Triage Interface

The Triage Interface serves as the primary interaction point between users and the AI system. It allows support personnel to input incoming messages and instantly receive AI-driven insights, structured classifications, extracted entities, and suggested responses.

This interface significantly reduces manual effort by automating the initial screening of messages. The integration of Large Language Models ensures that messages are analyzed contextually rather than simply through keyword matching.

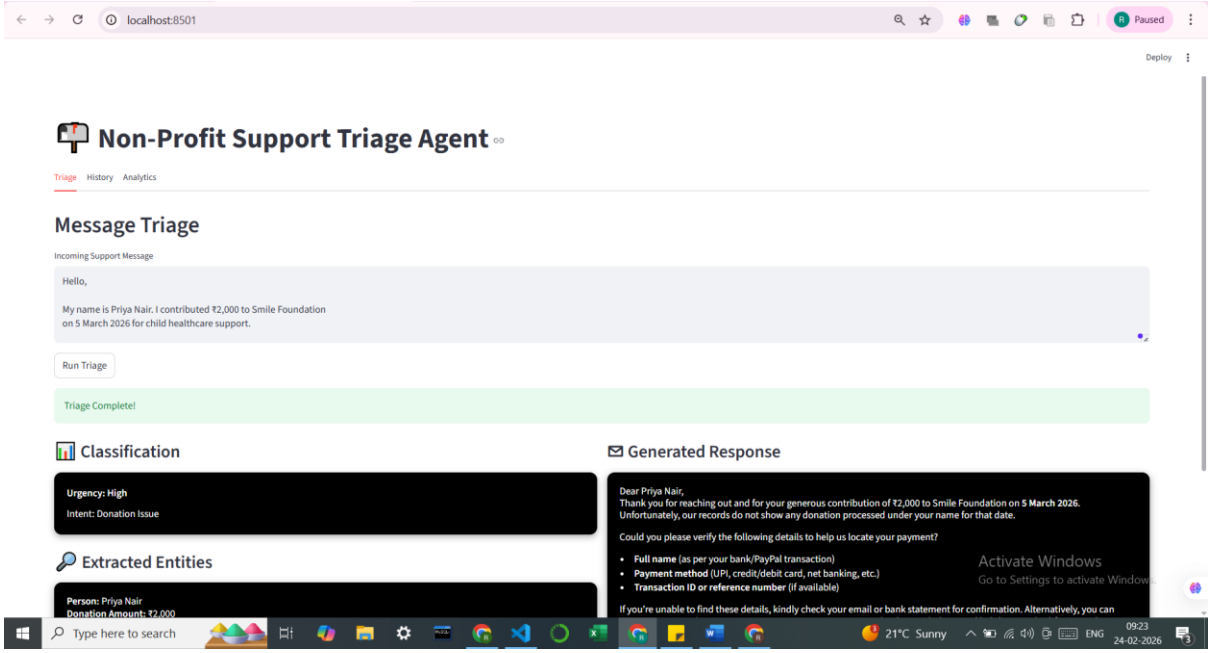
The triage interface performs multiple functions simultaneously:

- Accepts incoming text messages.
- Processes data through AI agents.
- Displays structured classification outputs.
- Generates contextual responses in real time.
- Provides summarized message insights.

The interface visually presents results through intuitive UI elements such as classification badges, entity displays, and formatted response sections. These visual indicators improve usability and reduce cognitive load for support personnel.

Component	Purpose
Message Input Field	Allows user message entry
AI Processing Indicator	Shows system working status
Classification Display	Shows urgency and intent
Entity Extraction Panel	Displays extracted information
Generated Response Section	Provides AI suggested reply
Summary Box	Quick overview of message

Table 4.1: Functional Components of Triage Interface



4.2 History Dashboard

The History Dashboard stores and displays previously processed messages along with their AI-generated insights. This feature is essential for operational transparency, auditing, and longitudinal analysis of communication trends.

By maintaining structured historical records, organizations can:

- Track recurring issues
- Monitor support workload
- Review past interactions
- Identify communication patterns

The history dashboard provides a card-based visualization approach, making it easy for users to scan past cases quickly.

Each case card typically includes:

- Message summary
- Urgency classification indicator
- Intent classification tag
- Extracted entity count
- Search and filtering options

This structured representation enhances usability while ensuring important details remain accessible.

Importance of History Dashboard

The history dashboard is particularly useful for:

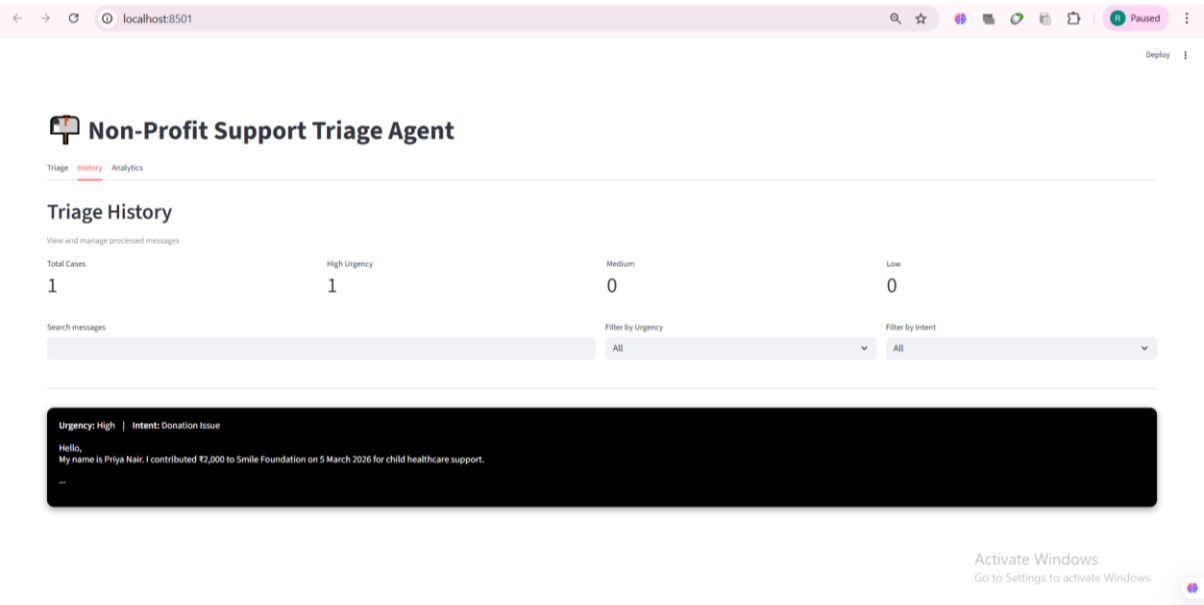
- Quality assurance audits
- Identifying unresolved cases
- Monitoring support trends
- Knowledge base development

Additionally, the integration of search and filter capabilities allows users to retrieve specific cases efficiently.

Feature	Benefit
Message Summary	Quick understanding of case
Urgency Indicators	Prioritization visibility
Intent Tags	Categorization clarity

Search Functionality	Fast retrieval
Filtering Options	Customized analysis

Table 4.2: History Dashboard Features



4.3 Analytics Dashboard

The Analytics Dashboard transforms processed message data into strategic operational insights. It provides visualizations that help organizations understand communication trends, workload distribution, and recurring issue categories.

Analytics dashboards are essential for:

- Decision-making
- Resource allocation
- Performance evaluation
- Predictive planning

The dashboard includes several key analytical components.

Cases by Urgency

This visualization shows how many cases fall into high, medium, and low urgency categories.

Significance:

- Detects operational pressure points
- Helps allocate resources effectively
- Identifies critical support periods

Cases by Intent

This chart highlights the most frequent types of user requests, such as donation issues or volunteer inquiries.

Significance:

- Reveals common organizational challenges
- Supports targeted process improvements
- Helps identify communication trends

Trend Analysis

Trend charts display message volume over time, helping organizations understand communication growth patterns.

Significance:

- Predicts workload increases
- Assists staffing decisions
- Enables proactive support planning

Distribution Metrics

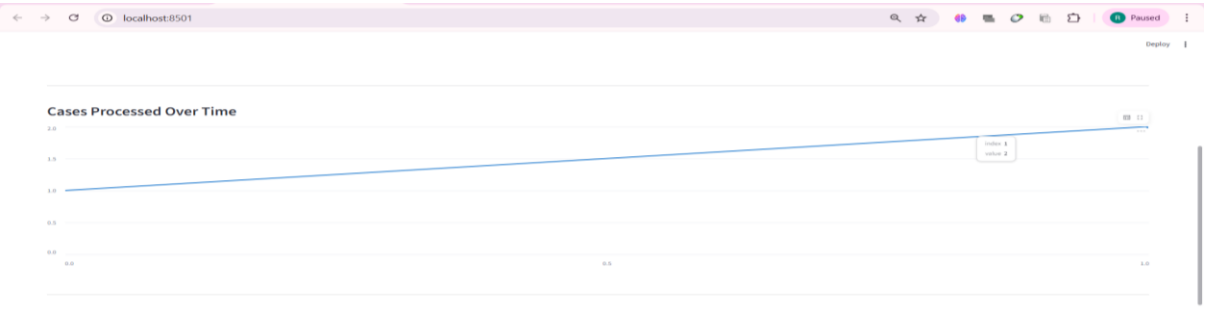
These metrics provide a comprehensive overview of message distribution across categories.

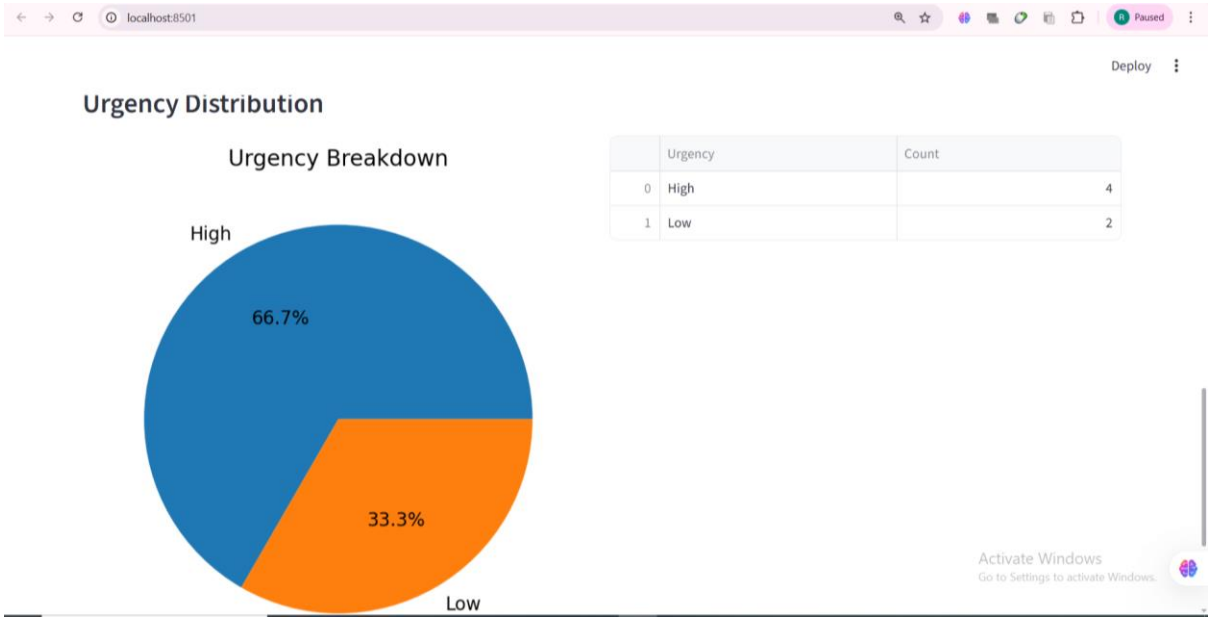
Significance:

- Holistic operational view
- Strategic decision support
- Performance monitoring

Metric	Purpose	Impact
Urgency Distribution	Priority monitoring	Faster response
Intent Analysis	Issue categorization	Process improvement
Trend Analysis	Time-based insights	Planning support
Distribution Metrics	Overall performance	Strategic decisions

Table 4.3 : History Dashboard Features




localhost:8501
Paused
Deploy
Urgency Distribution
Urgency Breakdown
High
66.7%
33.3%
Low
Urgency
Count
0
High
4
1
Low
2
Activate Windows
Go to Settings to activate Windows.

5. Results & Output Analysis

5.1 Sample Output Example

To evaluate the performance and effectiveness of the AI-powered Non-Profit Support Triage Agent, a representative sample message was processed through the complete AI pipeline. This evaluation demonstrates how the system handles real-world communication scenarios by performing classification, entity extraction, and contextual response analysis simultaneously.

Sample Input Message

"I donated $200 yesterday but the payment failed. Please contact me at sarah@email.com."

This message represents a typical donor communication scenario involving a payment issue. Such messages often require prompt attention since they directly impact donor trust and financial transactions.

System Output

After processing the message through the AI pipeline, the system generated the following structured output:

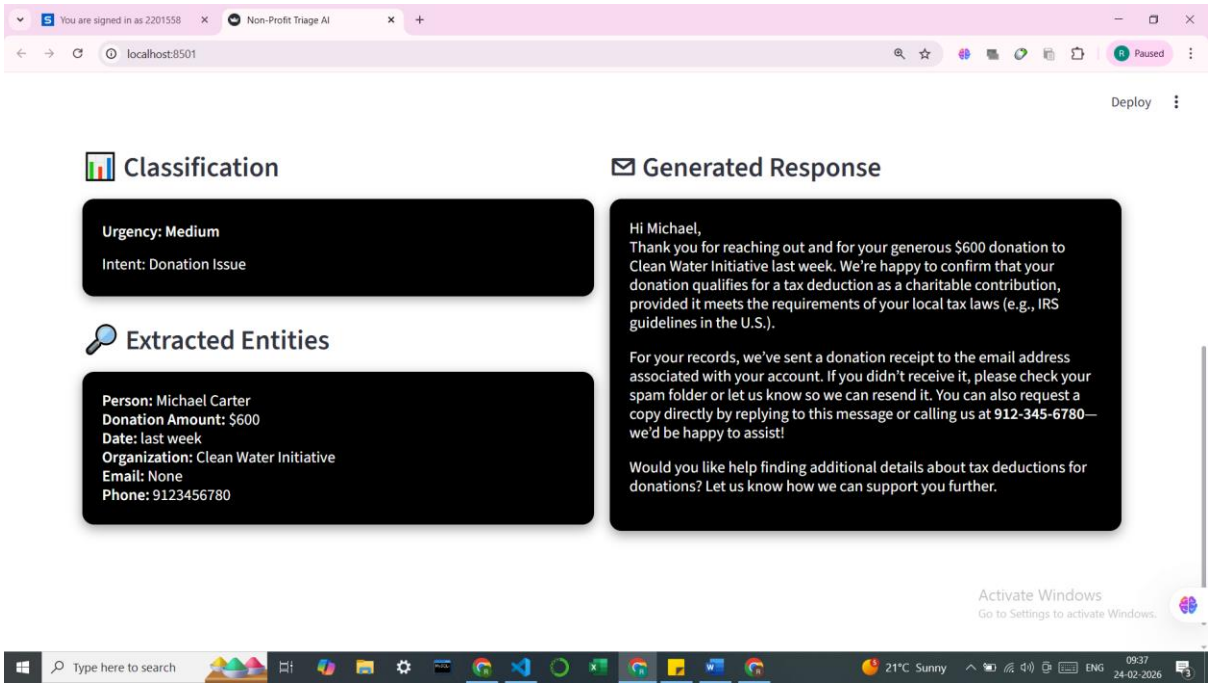
- **Urgency:** High
- **Intent:** Donation Issue
- **Extracted Entities:**
  - Name: Sarah
  - Amount: $200
  - Email: sarah@email.com

The system successfully classified the urgency as high due to the presence of a donation payment failure, which is typically a critical operational issue. Intent detection accurately categorized the message under donation-related concerns. Additionally, the Named Entity Recognition module extracted relevant structured information, including donor name, donation amount, and contact email.

This output demonstrates several important system capabilities:

- Contextual understanding of natural language input
- Accurate urgency classification based on message semantics
- Extraction of structured data from unstructured text
- Generation of actionable operational insights

Such structured intelligence significantly enhances operational efficiency by allowing support teams to respond faster, prioritize critical cases, and maintain accurate records without manual intervention.



5.2 Key Outcomes

The implementation of the AI-powered triage system produced several measurable operational and technical outcomes. These outcomes reflect improvements in communication efficiency, data management, and technological integration.

Operational Outcomes

The operational benefits observed include:

- Automated classification of incoming messages
- Reduction in manual message processing workload
- Improved response consistency across communications
- Faster prioritization of urgent cases
- Enhanced donor and volunteer communication quality

Automation ensures that repetitive tasks such as message reading, categorization, and initial response drafting are handled efficiently by AI systems, allowing human staff to focus on more complex and sensitive cases.

Technical Outcomes

From a technical perspective, the project achieved the following milestones:

- Successful integration of Large Language Models for classification and response generation
- Functional deployment of ChromaDB vector database for semantic storage
- Implementation of an interactive Streamlit-based analytics dashboard
- Development of a modular, scalable agent-based architecture

These achievements demonstrate the practical feasibility of combining modern AI technologies into a unified operational intelligence system.

Category	Outcome
Automation	AI-based message triage
Intelligence	Structured data extraction
Analytics	Dashboard visualization
Scalability	Modular AI architecture
Efficiency	Reduced response time
Accuracy	Improved classification consistency

Table 5.1: Project Outcomes

Chapter 6: Evaluation & Impact

6.1 Operational Impact

The deployment of the AI-powered triage system has significant operational implications for non-profit organizations. By automating initial message processing, organizations can achieve faster response cycles, improved communication accuracy, and better resource allocation.

Key operational impacts include:

- **Faster Response Times:** Automated classification ensures urgent cases are identified immediately.
- **Improved Prioritization Accuracy:** AI-based analysis reduces subjective human interpretation errors.
- **Enhanced Donor Communication:** Personalized responses improve donor satisfaction and engagement.
- **Better Resource Allocation:** Support teams can focus on critical cases instead of routine classification tasks.

Automation also improves organizational transparency by maintaining structured historical records of communications, enabling performance monitoring and strategic planning.

6.2 Technical Impact

From a technological standpoint, the project demonstrates the successful integration of advanced AI methodologies into a real-world operational application.

Technical contributions include:

- Practical deployment of Large Language Models in operational workflows
- Implementation of agent-based AI architecture for modular processing
- Integration of vector databases for semantic search capabilities
- Application of advanced prompt engineering techniques

These implementations highlight the feasibility of AI-driven operational analytics and demonstrate how emerging AI technologies can be applied beyond experimental research into practical organizational solutions.

Chapter 7: Conclusion

The AI-Powered Non-Profit Support Triage Agent represents a comprehensive solution for automating communication management in non-profit organizations. By combining Natural Language Processing techniques, Large Language Models, vector database storage, and interactive dashboard analytics, the system addresses critical operational challenges related to communication efficiency and data utilization.

The system successfully demonstrates:

- Automated urgency and intent classification
- Accurate structured entity extraction
- Context-aware response generation
- Semantic message storage using vector embeddings
- Real-time operational analytics visualization

This project effectively bridges the gap between theoretical AI concepts and practical implementation. It illustrates how AI technologies can enhance organizational workflows, improve communication quality, and support data-driven decision-making in non-profit environments.

Moreover, the modular architecture ensures scalability, allowing future integration of additional AI capabilities such as sentiment analysis, multilingual processing, and predictive analytics.

Overall, the project highlights the growing importance of Artificial Intelligence in operational intelligence systems and showcases its potential to transform communication management practices in socially impactful organizations.

Chapter 8: Future Scope & Enhancements

While the current implementation demonstrates significant capabilities, several enhancements can further improve the system's performance, scalability, and real-world applicability.

Potential Future Developments

1. Email Communication Integration

Integration with email APIs would allow real-time processing of incoming organizational communications. This would automate message triage directly from email inboxes, improving operational efficiency.

2. CRM Platform Connectivity

Connecting the system with Customer Relationship Management platforms would enable centralized donor and volunteer management. This integration would improve data consistency and organizational workflow coordination.

3. Multilingual Support

Expanding the system to support multiple languages would enhance accessibility for global non-profit organizations. Multilingual NLP capabilities would enable broader adoption.

4. Sentiment Analysis Integration

Adding sentiment analysis would allow detection of emotional tone in messages. This could help prioritize emotionally sensitive communications such as complaints or crisis requests.

5. Automated Escalation Mechanisms

Critical cases could be automatically escalated to appropriate departments or personnel based on urgency classification, ensuring faster resolution.

6. Retrieval-Augmented Generation (RAG)

Integration of Retrieval-Augmented Generation would enhance response accuracy by combining stored knowledge with LLM generation capabilities.

7. Cloud Deployment for Scalability

Deploying the system on cloud platforms would ensure scalability, reliability, and accessibility for larger organizations.

8. Advanced Analytics Dashboards

Future dashboards could include predictive analytics, communication trend forecasting, and performance monitoring metrics.